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EXERCISE SHEET / THEME 7: STANDARDIZATION AND THE NORMAL DISTRIBUTION.

EXERCISE 1. PERCENTILES AND OLDER ADULTS.

Context: You work for your country's national service for older adults. The government is considering launching a programme for dependent older adults. Your team has conducted a representative survey of 500 older adults. One of your team's analysts has prepared the following table, which indicates the level of dependency among the surveyed population.

Level of dependence	Frequency	Percentage	Cum. percentage
Self-sufficient	220	44	44
Mild Dependence	120	24	68
Moderate Dependence	90	18	86
Severe Dependence	70	14	100
Total	**500**	**100**	

Answer the following questions:

a) According to the survey, what level of dependency does the 50th percentile of the older adult population have?
b) According to the survey, what level of dependency does 70% of the older adult population have?
c) The government does not have sufficient funds to implement a universal program. Suppose you are going to propose covering the 20% of older adults with the highest levels of dependency. What levels of dependency would be included under your proposal?

EXERCISE 2. NORMAL DISTRIBUTION AND PRIMARY CARE WAITING TIMES.

Context: You are analyzing waiting times in the health system. The waiting time for primary care patients is normally distributed with a mean of 12 weeks and a standard deviation of 3 weeks.

Answer the following questions:

a) What is the probability that a person waits between 9 and 15 weeks?
b) What is the probability that a person waits between 6 and 18 weeks?
c) The Ministry of Health is dealing with budget issues. Even though 12 weeks is a long time for primary care attention, they will focus on the worst-off 25% of the population (i.e. the 25% who wait the longest). From how many weeks onward is a person considered part of the worst 25%?
d) As a next step, the government would like to cover the remaining population. However, they believe that waiting 4 weeks is not a problem, so they will not consider them for this second step. What percentage of the total population will be covered in this step? Hint: remember to exclude also those covered in the first step (part c).